

ORDERING OF DIPOLES IN DIFFERENT TYPES OF MICROTUBULE LATTICE

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Microtubules (MTs) are the largest protein polymers in the cytoskeleton of eucaryotic cells in which they perform various functions. They are important in cell division, cell movement, they seem to be the devices through which are transferred signals in the nervous system. In this paper we continued to investigate the hypothesis that MTs can be viewed as assemblies of dipoles which are carried by the MT subunits, tubulin heterodimers. These assemblies were studied by means of the two-dimensional Ising model for both the A- and B-type arrangements of the tubulin subunits in a MT and the number of protofilaments 12, 13 and 14. We found that depending on these parameters and the magnitudes and orientations of the dipoles a MT may be at body temperature in an ordered phase or in a phase characterized by a random configuration of dipoles. The type of the ordered phase is determined by the above parameters as well, and it can be ferroelectric, antiferroelectric or ferrielectric. The dipolar ordering also depends on the presence of microtubule associated proteins, assuming that they can locally alter the dipolar interactions by binding to a MT, and external electric fields. The model presented here started by Tuszyński^{1–3} can be one of the first steps in the theoretical investigation of the electromagnetic features of MTs and their role in the MT behavior.

1. Introduction

Microtubules (MTs) are structural elements of eucaryotic cells.⁴ They are hollow tubes formed by the polymerization of tubulin molecules. Each tubulin subunit is a dimer composed of two monomers termed α - and β -tubulin.^{5–7} The dimer is approximately 8×10^{-9} m long and its width is about 5×10^{-9} m. The molecules of tubulin bound within a MT arrange themselves into protofilaments. The majority of MTs *in vivo* have 13 protofilaments.^{8,9} MTs assembled *in vitro* consist mostly of 13 and 14 protofilaments. Changes in the number of protofilaments occur also within a single MT, even though rather infrequently.⁹

Almost all MTs inside the cell form an array around the centrosome.⁴ In a non-dividing cell these MTs continuously grow and shrink due to the constant addition and dissociation of tubulin molecules at both MT ends. This phenomenon is known as dynamic instability¹⁰ and its microscopic mechanism is largely unexplained. It is utilized in cell division when MTs growing in random directions from two centro-

somes positioned at the opposite poles of the cell eventually attach to chromosomes in order to pull an equal number of them to each of the cell extremities. In a stable state MTs serve as tracks for the transport of organelles. This transport is mediated by motor proteins which attach to MTs.

Stable MTs form parallel networks in dendrites and axons of neurons. These networks have been compared to parallel information processors that may be basic for the functioning of the nervous system.¹¹ Nondynamic MTs also form the cores of cilia and flagella that extend from the cell surface and move in a coordinated way. It is conjectured that such motion is a result of the cooperative interactions between the ciliary motor proteins.

The physical pathways of the MT activities briefly outlined above have been a subject of growing theoretical investigation. Several researchers have suggested that the energy freed in the GTP hydrolysis which takes place soon after the addition of each tubulin molecule to a MT end propagates along a MT as a solitary wave.^{1,12-14} This wave may locally alter the elastic state of tubulin which may consequently lead to a dissociation of a tubulin dimer from a MT end or may be involved in the coordinated behavior of the motor proteins.

The present paper is focused on the investigation of the dipolar order in MTs. This question has been studied by Tuszyński *et al.* who put forward the hypothesis that some aspects of the MT behavior may be due to the interactions between the permanent dipoles carried by the tubulin molecules.² So far Tuszyński and his coworkers have investigated the dipolar ordering only in MTs that are composed of 13 protofilaments and are represented by the A lattice (see the following section).³ In this work, a similar type of analysis is extended to MTs with both A and B lattices and number of protofilaments 12, 13 and 14.

2. Microtubule Lattice

Neighboring MT protofilaments are longitudinally shifted with respect to each other. Consequently, the tubulin dimers bound in a MT are arranged in a hexagonal manner and a MT can be represented by a hexagonal lattice.

As mentioned earlier, the most common number of protofilaments in a MT *in vivo* is 13 and *in vitro* 13 and 14 where MTs consisting of different numbers of protofilament in the range $p = 12-17$ have been also observed.¹⁵ In all MTs, independent of the number of protofilaments, the tubulin monomers are arranged along a 3-start helix, i.e. each MT consists of three helices of tubulin monomers.

Depending on the geometrical arrangement of the α and β monomers two types of the MT lattice can be defined. They have been termed the A lattice and the B lattice. In MTs that are represented by the A lattice the α and β monomers along a 3-start helix alternate. If the number of protofilaments is odd then the connection between two monomers which are positioned in the last and first protofilament along a 3-start helix is heterologous, i.e. it is an α - β or a β - α connection. Hence, MTs with these A lattices are helically symmetric [Fig. 1(a)]. In MTs that possess an A

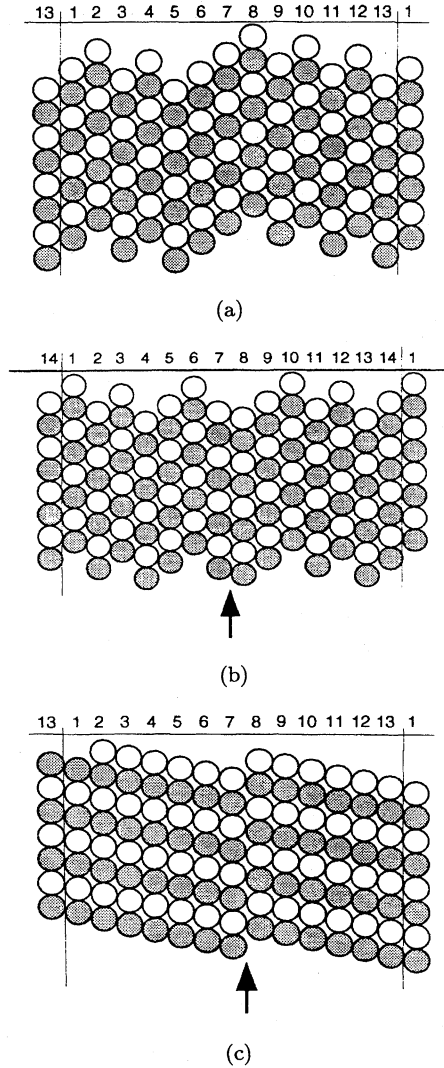


Fig. 1. A two-dimensional schematic view of a MT: (a) A lattice for $p = 13$, (b) A lattice for $p = 14$, (c) B lattice for $p = 13$. The α monomers are white, the β monomers are grey. The seam is indicated with an arrow. [Drawn according to (Martin *et al.*, 1993).]

lattice with an even number of protofilaments this connection is homologous, i.e. it is an α - α or a β - β connection. This means that in such MTs there is a discontinuity between the first and the last protofilament. This discontinuity is called a "seam" [Fig. 1(b)]. A seam is present in all MTs that possess a B lattice. In these MTs each of the three 3-start helices consists of sections that are composed of p monomers of one type. Along one 3-start helix each such section is connected at both ends to sections formed of p monomers of the opposite type. This results in heterologous monomer-monomer contacts along the seam in MTs with a B lattice [Fig. 1(c)].

To maintain the 3-start helix monomer symmetry of MTs that are composed of different numbers of protofilaments one can adjust the longitudinal shift between two neighboring protofilaments that is also referred to as stagger, as was done in Figs. 1(a) and 1(b). In this way the distances between the monomers and dimers are slightly changed so that the values of the stagger were calculated according to the expression $(4 + 12/p) \times 10^{-9}$ m for the A lattice and $(12/p) \times 10^{-9}$ m for the B lattice^a (see Fig. 4).

From the consideration above it is obvious that if each monomer in the MT was represented by a site, the A and B type arrangements of tubulin dimers would be represented by the same lattice.⁸ The fact that the A and B lattices are equivalent on the level of monomers may suggest that MTs possess only one of them. Recent experimental evidence indicates that the MT lattice is a B lattice.^{6,16,17} This result was obtained indirectly by decorating tubulin with motor protein kinesin since due to the very similar structure of α - and β -tubulin current observational techniques cannot satisfactorily detect the difference between the two tubulin monomers.^{6,8} However, at present no definite conclusions can be made about the type of the MT lattice.

3. A Model of Dipolar States of Tubulin and their Interactions in a Microtubule

Experimental investigations of Athenstaedt¹⁸ have shown that a number of biological structures including MTs exhibit pyroelectric and piezoelectric behavior. Such properties originate as a result of the elementary dipoles carried by the constituent particles of the structure. This implies that a tubulin dimer carries a permanent dipole moment. However, the charge distribution within the tubulin molecule is not very well known. It seems, though, that at one end of the tubulin heterodimer the density of ions sitting on the amino acid residues of the protein chains is higher than in the rest of the molecule.¹⁹ This attracts charges of the opposite polarity from the MT surroundings and can give rise to a dipole. What would be the charge and the length of the dipole is not precisely clear. Consequently, in the numerical calculations presented here these parameters were adjustable. As will be seen below the adopted value of the magnitude of the dipole moment was not unreasonable.

The model of dipolar interactions utilized in this paper was suggested by Tuszyński *et al.*¹⁻³ According to this model a MT is viewed as an assembly of dipoles that are placed at sites of a two-dimensional hexagonal MT lattice. In this hexagonal lattice each three nearest sites form identical triangles. Hence the MT lattice is also a triangular lattice. The directions along the sides of a triangular cell determine the three directions of the dipole-dipole interactions in the MT lattice (see Fig. 2).

^a An A or a B lattice that represents a MT which consists of p protofilaments will be also termed p -A, p -B, where p can be 12, 13 or 14.

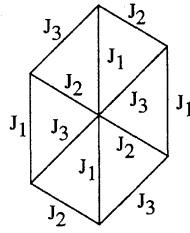


Fig. 2. A section of a MT lattice that represents a dipole and its six nearest neighbors. Each of the three directions along the sides of a triangular cell is associated with one of the energy constants J_1 , J_2 and J_3 .

In the model each tubulin dipole interacts only with its six nearest neighbors and it can be in one of two orientational states that are characterized by the same magnitude of the dipole moment. These states were called the α and the β state. (The α and the β state of a tubulin dimer originally refer to two electrostatic states of tubulin that have been adopted in the cellular automata models.¹¹ These states were associated with a mobile electron that could be localized either more toward the α or more toward the β monomer of the dimer.) Since the longitudinal axis of the tubulin dimers assembled in a MT is parallel to the MT axis we have assumed that in the two dipolar states the projections of the dipole on the MT axis point in opposite directions. This direction was chosen to be positive in the α state and negative in the β state. Hence, the two dipolar states can be also termed the up and down state.

First we considered the simplest case in which the dipole on the tubulin is parallel to the MT axis in both α and β state. If an array of such dipoles is placed at sites of a triangular lattice and it is subjected to an external electric field $\mathbf{E}$ the total potential energy of the whole system is

$$H = - \sum_{j=1}^{N_C} \sum_{i=1}^{N_R} (J_1 \sigma_{i,j} \sigma_{i+1,j} + J_2 \sigma_{i,j} \sigma_{i-1,j+1} + J_3 \sigma_{i,j} \sigma_{i,j+1}) - \mathbf{p} \cdot \mathbf{E} \sum_{j=1}^{N_C} \sum_{i=1}^{N_R} \sigma_{i,j}. \quad (1)$$

In the sum (1) N_R is the number of rows and N_C is the number of columns of the triangular lattice, $\sigma_{i,j}$ is a dipole variable that can have two values $+1$ or -1 which correspond to the two tubulin dipole states $\pm qs$ and J_1 , J_2 , J_3 are energy constants that characterize the interactions between the nearest dipoles along the three directions in the MT lattice (J_1 , J_2 , J_3 direction, Fig. 2). As can be seen Eq. (1) represents the Hamiltonian of a two-dimensional Ising model.²⁰

The interaction constants J_n were found from the expression

$$W_n^{\uparrow\uparrow} - W_n^{\uparrow\downarrow} = -2J_n, \quad (2)$$

where $W_n^{\uparrow\uparrow}$ is the interaction energy between two neighboring dipoles in the n th direction when they both are in the up state and $W_n^{\uparrow\downarrow}$ is the interaction energy between two neighboring dipoles in the n th direction when one dipole is up and the other dipole is down. $W_n^{\uparrow\uparrow}$ and $W_n^{\uparrow\downarrow}$ were calculated using the classical formula for the interaction energy between two dipoles

$$W_n = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{(\mathbf{p}_1 \cdot \mathbf{p}_2)_n - 3(\mathbf{u} \cdot \mathbf{p}_1)_n(\mathbf{u} \cdot \mathbf{p}_2)_n}{|\mathbf{x}_1 - \mathbf{x}_2|_n^3}. \quad (3)$$

Here $\mathbf{p}_1$ and $\mathbf{p}_2$ are the dipole moments of the dipoles whose components with respect to the MT axis in our case have the magnitudes $\pm qs$, where q is the charge of the dipole and s is its length, $\mathbf{u}$ is a unit vector oriented along a line that connects the centers of the two dipoles, $\mathbf{x}_1$, $\mathbf{x}_2$ are the position vectors of the centers of the dipoles, ϵ_0 is the permittivity of vacuum and ϵ_r is the relative permittivity of the medium.

Since the magnitude and the orientation of the tubulin dipole are not very clear we have also considered a more general case in which the dipole in one of the states, for example, β is tilted with respect to the protofilament axis (Fig. 3). Because a MT is a cylinder the orientation of the dipole in the β state is given by two angles. One angle gives the orientation with respect to the MT axis and the other angle with respect to an axis that is perpendicular to the MT surface and passes through the center of the dipole. In the calculations presented here the curvature of the MT was not taken into account. The MT was treated as a sheet and the dipoles were assumed to lie in the plane of the sheet so that the tilting in the β state was given only by one angle θ . As will be seen in Sec. 9 varying θ results in various low temperature configurations of dipoles in the MT as the interaction constants J_n 's change [see Eq. (3)]. Including the angle that characterizes the curvature would have the same effect. Hence to show the capabilities of the model it seemed to us sufficient to do calculations using the two-dimensional case at this moment.

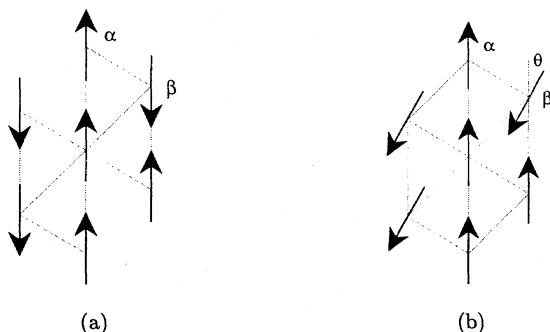


Fig. 3. A dipole and its nearest neighbors in a MT A lattice. (a) In both α and β state the dipole is parallel to the MT axis and oriented in opposite directions as is indicated by arrows. (b) In the β state the dipole is tilted by an angle θ with respect to the negative MT axis.

In the MT lattice in which the dipoles are tilted in the β state and lie in a two-dimensional plane the potential energy due to the interaction between the dipoles can be given by the following expression (Yeomans, 1992)

$$H_t = \frac{1}{4} \sum_{n=1}^3 \sum_{\langle NN \rangle} J_{n1}(1+s_k)(1+s_l) + J_{n2}(1-s_k)(1-s_l) + J_{n3}[(1+s_k)(1-s_l) + (1-s_k)(1+s_l)]. \quad (4)$$

In the equation above, the sum over n runs over the three sides of each triangular cell and $\langle NN \rangle$ denotes the sum over the nearest neighbors. The variable s_k can have two values $+1$ or -1 which correspond to the α and β state, respectively. Each direction n of the triangular cell is characterized by three interaction energies: J_{n1} is the interaction energy between two dipoles that are in the nontilted state, J_{n2} is the interaction energy between two dipoles that are both tilted and J_{n3} is the interaction energy between a dipole in the tilted and a dipole in the nontilted state.

The terms in Eq. (4) can be rearranged so that one can write it in the form

$$H_t = H_{\text{int}} + V_{\text{eff}} \sum_{j=1}^{N_C} \sum_{i=1}^{N_R} \sigma_{i,j} + C, \quad (5)$$

where H_{int} is the same as the interaction part of (1) but with a plus sign, V_{eff} is an effective potential energy, and C is a constant. The three interaction constants J_n in H_{int} are given by the equations

$$\begin{aligned} J_1 &= \frac{1}{4}(J_{11} + J_{12} - 2J_{13}), \\ J_2 &= \frac{1}{4}(J_{21} + J_{22} - 2J_{23}), \\ J_3 &= \frac{1}{4}(J_{31} + J_{32} - 2J_{33}), \end{aligned} \quad (6)$$

where the interaction energies J_{nm} were calculated from the formula (3). The effective potential energy V_{eff} and the constant term C have the following form

$$V_{\text{eff}} = \frac{1}{2}(J_{11} - J_{12} + J_{21} - J_{22} + J_{31} - J_{32}) \quad (7)$$

and

$$C = \frac{N}{4} \times (J_{11} + J_{12} + 2J_{13} + J_{21} + J_{22} + 2J_{23} + J_{31} + J_{32} + 2J_{33}). \quad (8)$$

In the latter equation $N = N_C \times N_R$ is the total number of lattice sites.

In Eqs. (1) and (5) toroidal boundary conditions have been used. Due to the helical arrangement of the tubulin dimers a dipole in the last protofilament, σ_{i,N_C} ,

is neighboring to the dipole $\sigma_{i+a,1}$, where a depends on the stagger. The formulas (6)–(8) are the same both inside and at the boundaries of the A lattice with an odd number of protofilaments. All other lattices contain a seam at which the distances between the tubulin dimers are different and, consequently, the values of the interaction constants are different from those inside the lattice. This has to be taken into consideration in the three equations above.

It is not difficult to show from Eq. (3) that the values of the interaction constants J_n and the interaction energies J_{nm} depend on the type of the MT lattice, i.e. on the stagger which determines the distances $|(\mathbf{x}_1 - \mathbf{x}_2)_n| = r_n$ (Table 1) and therefore the angles θ_n (Fig. 4), on the orientation of the tubulin dipole when it is tilted and on the parameters q , s and ϵ_r . The dependence on the latter three parameters can be reduced to a dependence on the single parameter $\bar{Q} = q^2 s^2 / \epsilon_r$. In a realistic model, q should have a value of the order of one electronic charge 1.602×10^{-19} C and s should be of the order of several nanometers. As a relative permittivity one can use the value measured for proteins. This value is between 20 and 30 and it is

Table 1. Staggers and distances r_2 and r_3 inside the A and B lattices. The parameter $r_1 = 8 \times 10^{-9}$ m (length of the tubulin dimer).

A LATTICE			
p	12	13	14
stagger (10^{-9} m)	5.00	4.92	4.86
r_2 (10^{-9} m)	5.83	5.87	5.90
r_3 (10^{-9} m)	7.07	7.02	6.97

B LATTICE			
p	12	13	14
stagger (10^{-9} m)	1.00	0.92	0.86
r_2 (10^{-9} m)	8.60	8.67	8.72
r_3 (10^{-9} m)	5.10	5.08	5.07

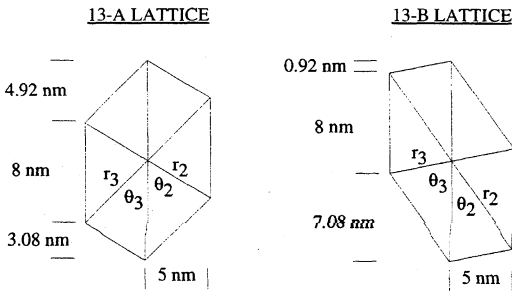


Fig. 4. The geometrical parameters of the 13-A and inside the 13-B lattice. The lateral separation between two neighboring protofilaments was taken as 5×10^{-9} m (Rasmussen *et al.*, 1990).

the effective relative permittivity, i.e. such which reflects the response of both the protein and its solvent.²¹ We can point out that this value is not so much different from the value used in the previous papers by Tuszyński *et al.* This was $\epsilon_r = 70$ which is the value for free water in the static case (for nonoscillating dipoles) at body temperature.²²

4. Numerical Procedure: Monte Carlo Method

The numerical calculations presented in this work have been performed on the basis of the classical Monte Carlo method that was realized by means of the Metropolis algorithm.²³ The procedure was used to find the configurations of the interacting system of dipoles in a MT at different temperatures. The configuration is a result of the competition between the potential energy of the dipoles and thermal energy. The interactions with neighboring dipoles and an electric field force a dipole to assume a state with the lowest potential energy (ground state). If the thermal energy is small compared to the potential energy the probability that the dipole will change its state due to a random thermal fluctuation is very small. With increasing thermal energy this probability also increases as the charges of the dipoles become more free to move.

In the algorithm the change of state of a dipole (a flip) happens when the number

$$\exp \left[- \frac{(E_{\text{new}} - E_{\text{old}})}{k_B T} \right], \quad (9)$$

is greater than a random number with a value between 0 and 1. In (9) E_{old} is the energy of a dipole in its old state, E_{new} is the energy of the dipole in its possible new state and k_B is Boltzmann's constant. Since in our model each dipole interacts only with its nearest neighbors the two energies depend only on the configuration of these dipoles.

It is obvious that a dipole always flips if $E_{\text{new}} < E_{\text{old}}$ because in that case the argument of the exponential in (9) is a positive number. In the opposite situation the argument is negative and the dipole flips only if (9) is larger than a randomly generated number whose value is between 0 and 1. This means that the probability of flipping a dipole is larger at higher temperatures. Such a procedure ensures that locally configurations of dipoles can form that are not in a ground state. With the increasing temperature the number of these regions increases and eventually at high temperatures the whole array of dipoles in a MT is completely randomized.

In the program used for the calculations shown below the number (9) was calculated and compared to the random number at each lattice site. The lattice was examined column by column. One such examination of the lattice is one Monte Carlo step. The dipolar ordering was calculated at different temperatures in the range 1–2000 K. The temperature step varied from $\Delta T = 1$ K to $\Delta T = 5$ K. In order for the array of dipoles to become uncorrelated with its initial state, at each investigated temperature 1000 Monte Carlo steps were performed and the last 200

were used to calculate polarization per site (relative polarization). If the simulation was started from the low temperature end we chose as the starting configuration one of the ground states of the system. In the simulations started from the high temperature end the configuration at the beginning of the simulation was a random configuration. The final configuration from the previous Monte Carlo step was used as the initial configuration for the next Monte Carlo step.

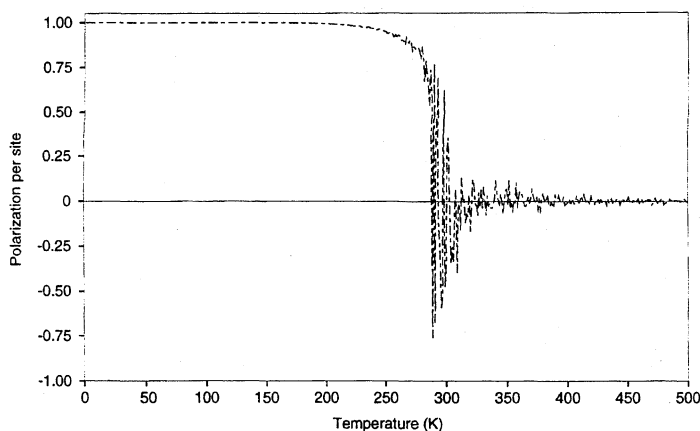
Polarization per site was calculated after each Monte Carlo step (starting from the step 801) by dividing the sum of the values of the dipole variables $\sigma_{i,j}$ at all lattice sites by the total number of lattice sites. A dipole tilted in the β state contributed to the sum by $(-\cos\theta)$. The values of relative polarization plotted in figures below are the average of the values calculated in the Monte Carlo steps 801 to 1000. All Monte Carlo simulation shown here were performed for a 100 rows long MT (with the exception of simulations that involved MAPs, see Sec. 8) and $\bar{Q} = 12 \times 10^{-56} \text{ C}^2\text{m}^2$. The polarization curves and data in tables in Secs. 5–8 were obtained assuming that the dipole in the β state is nontilted. The calculations for the case when the dipole in the β state is tilted are discussed in Sec. 9.

5. Microtubule Represented by the A Lattice

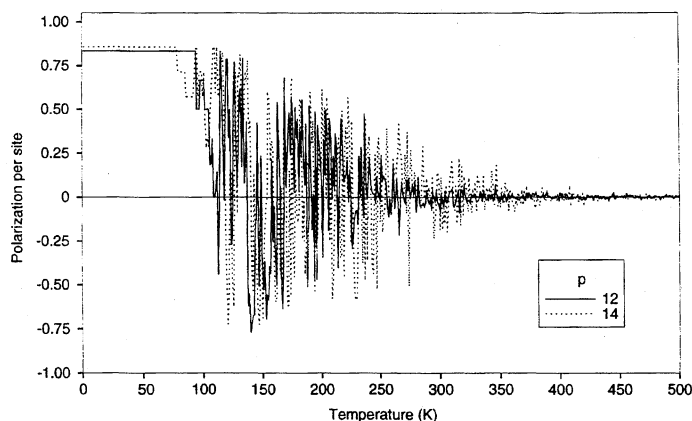
Figure 5(a) is a plot of polarization per site as a function of temperature calculated for a MT represented by the 13-A lattice. As can be seen at low temperatures the MT is in a completely ordered state with all dipoles pointing in the positive direction along the MT axis as was chosen at the beginning of the simulation. At temperatures close to the room temperature (300 K) the system undergoes a transition into a disordered state when the dipoles are oriented randomly and the total polarization is zero. The small fluctuations around zero are due to the thermal fluctuations. In analogy with ferroelectric crystals the two states of the array of dipoles in a MT will be further termed a ferroelectric and a paraelectric phase, respectively.²⁴

It is not difficult to explain the character of the curve shown in Fig. 5(a). By examining the values of the corresponding interaction constants that are listed in Table 2 it becomes obvious that all J_n 's are of the order of magnitude 10^{-21} J . The transition from the low temperature phase to the high temperature phase takes place when the energy of the thermal fluctuations $k_B T$ is sufficiently large to allow the dipoles to exist in configurations that do not correspond to one of the ground states. This happens when $k_B T$ is comparable to the interaction energy between the dipoles.²⁵ Since the value of the Boltzmann's constant is $1.38 \times 10^{-23} \text{ JK}^{-1}$ the transition occurs at temperatures $T \sim 100 \text{ K}$.

The considerations above imply that increasing the interaction energies between the dipoles in the MT increases the transition temperature and *vice versa* which we also verified by the Monte Carlo simulations. As was noted in Sec. 3, the magnitudes of the interaction constants J_n depend on the value of the parameter $\bar{Q} = q^2 s^2 / \epsilon_r$ and the geometrical configuration of the tubulin dimers in the MT which is given



(a)



(b)

Fig. 5. Polarization per site: (a) 13-A MT, (b) 12-A and 14-A MT.

by the type of the MT lattice A or B and the number of protofilaments. Changing these parameters changes the transition temperature and, consequently, determines the character of the dipolar order in a MT at body temperature (310 K).

Curves plotted in Fig. 5(b) represent polarization per site calculated for a 14-A and 12-A MT. The figure shows that at low temperatures in the 12- and 14-protofilament MTs not all dipoles point in the same direction — the value of the total relative polarization is less than one. This is due to the presence of the seam. As can be found in Table 2 in the 12-A and 14-A lattices the constant J_{3s} is negative and has a larger magnitude than any other interaction constant. The second most dominant interaction constant is J_1 which is positive and characterizes the dipole-dipole interaction along the longitudinal direction in the MT lattice.

Table 2. The interaction constants in the MT A lattice for $p = 12, 13, 14$. J_1, J_2, J_3 characterize the interaction inside the lattice, J_{2s} and J_{3s} refer to the seam.

p	12	13	14
J_1 (10^{-21} J)	+4.21	+4.21	+4.21
J_2 (10^{-21} J)	-1.12	-0.94	-0.79
J_3 (10^{-21} J)	+1.53	+1.49	+1.45
J_{2s} (10^{-21} J)	+1.67	-0.94	+1.65
J_{3s} (10^{-21} J)	-7.20	+1.49	-7.56

The form of the Hamiltonian (1) is such that when the interaction between two nearest dipoles is characterized by a negative interaction constant, this bond is in the state with the lowest energy if the dipoles point in the opposite directions. On the contrary, interaction between two nearest dipoles characterized by a positive interaction constant requires the dipoles to be aligned. Thus, the dipoles in the two protofilaments along the seam are at low temperatures aligned but in a different direction in each protofilament giving the total relative polarizations $10/12 = 0.833$ and $12/14 = 0.857$ for 12-A and 14-A lattices, respectively.

In a MT that is assembled from 13 protofilaments such a situation does not occur. This MT does not have a seam and the two dominant interaction constants J_1 and J_3 are positive. Consequently, even though the energetically more favorable state along the J_2 direction is when each two nearest dipoles point in the opposite directions, the effect of J_1 and J_3 overcomes the effect of J_2 and all dipoles at low temperatures are in the same state.

It can also be noticed that the interval of temperatures over which the transition from the ordered to the disordered state in the 12-A and 14-A lattices takes place is much larger than in the 13-A lattice. This effect is also caused by the presence of the seam. Since the magnitude of the interaction constant J_{3s} is larger than the magnitudes of the remaining four J_n 's, the energy of the thermal fluctuations must be sufficiently large to eventually overcome also the interaction through J_{3s} that forces the dipoles to be ordered along the seam in opposite directions which affects the dipolar ordering in the whole lattice.

Another feature of the array of dipoles with the potential energy (1) is that the state in which all dipoles are oriented in one or the other direction along the MT axis has the same energy. Due to this the ground state of the 13-A lattice is doubly degenerate and the 12-A and 14-A lattices possess a multitude of ground states. For example, all states in which the assembly of dipoles in the MT forms two subclusters, one consisting of protofilaments 1 to $N_T < N_C$ and all dipoles in it are up/down and the other of protofilaments $(N_T + 1)$ to N_C with all dipoles pointing down/up, have the same energy. For instance, the simulation for the 14-A lattice in Fig. 5(b) was started from the configuration in which $N_T = 1$ that is shown in Fig. 6(a).^b

^bIn all figures below that depict configurations of dipoles are included columns 0 and $(N_C + 1)$.

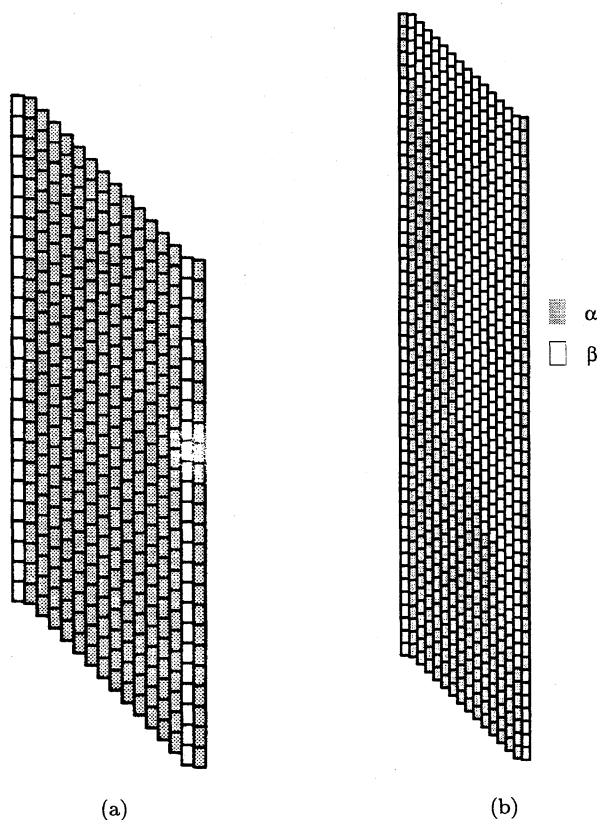


Fig. 6. (a) One of the ground states of a 14-A MT in which in the protofilaments 1 and 14 along the seam the dipoles are aligned in different directions and in the rest of the MT they point up. (b) A configuration of dipoles in a 50 row long section of the 14-A MT at $T = 5$ K. In the whole lattice (100 rows) the two subclusters in which the directions polarization are opposite have almost the same size so that the total relative polarization is almost zero.

The two clusters with different directions of polarization can also have a different form, for example, such as in Fig. 6(b). The figure shows a configuration of dipoles in a section of the 14-A lattice at temperature $T = 5$ K at which the system of dipoles arrived when it was initially completely randomized at $T = 1000$ K. This configuration is not a groundstate because in the 14-A lattice in a groundstate all dipoles in one protofilament have to point in the same direction. However, the total interaction energy is only slightly higher than the energy in a groundstate.

The polarization curve corresponding to the configuration shown in Fig. 6(b) is the solid line in Fig. 7. The dotted line in Fig. 7 represents a simulation that was also started from a random configuration of dipoles but the generated sequence of random numbers used by the Monte Carlo algorithm in this calculation was different. In this case in the ground state all dipoles in protofilaments 1 to 8 pointed up and 9 to 14 down so that the corresponding polarization was $2/14 = 0.143$.

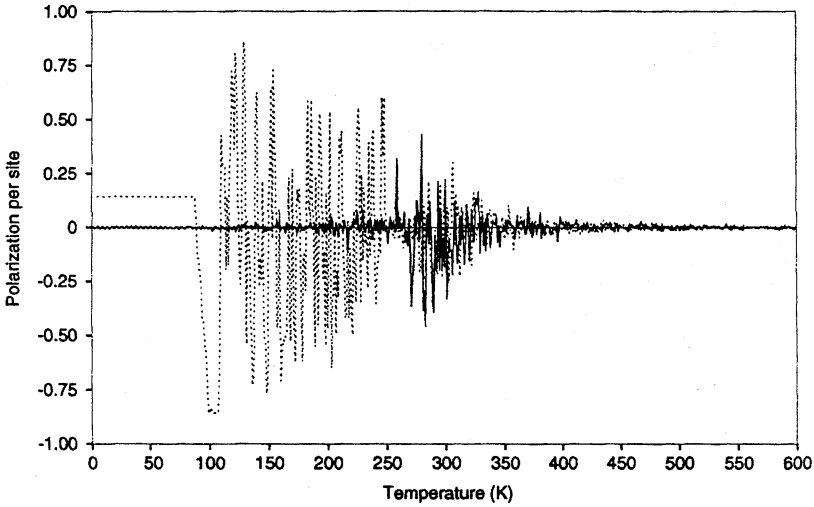


Fig. 7. Polarization curves calculated for a 14-A MT using two different sequences of random numbers. The simulations were started from random configurations of dipoles at $T = 1000$ K.

We can conclude that the presence of a seam in a MT with the A-lattice introduces a multitude of states that are characterized by different configurations of dipoles but the same energy. The ground state that the regular array of dipoles in a MT reaches in a Monte Carlo simulation depends on the initial configuration of the dipoles and the chosen sequence of random numbers. If the configuration of dipoles in a MT is changed by some external factor (for example, an external electric field) a MT can switch into a different ground state. At low temperatures the MT remains in the ground state for a long time. At high temperatures it is easier for a MT to switch between various states due to the increase in thermal energy. [According to Figs. 5(b) and 7 at temperatures from the range of about 0–100 K the polarization in the 12-A and 14-A lattice has the same value. Thus, a thermal fluctuation by several degrees does not alter the value of polarization. However, at higher temperatures the polarization can fluctuate within 1 K.]

6. Microtubule Represented by the B Lattice

As was explained earlier for any number of protofilaments a MT represented by the B lattice contains a seam along which the values of the interaction constants are different from those inside the MT (Table 3). Because the largest interaction constant in the B lattices is J_3 and the second largest J_1 the ground state of a MT that exhibits a B lattice is antiferroelectric: all dipoles in one protofilament point in the same direction and in each two neighboring protofilaments these directions are opposite. Since J_{2s} and J_{3s} are several times smaller than J_1 and J_3 the dipolar order along the seam will also be determined by the latter interaction constants.

Table 3. The interaction constants in the MT B lattice.

p	12	13	14
J_1 (10^{-21} J)	+4.21	+4.21	+4.21
J_2 (10^{-21} J)	+1.67	+1.66	+1.65
J_3 (10^{-21} J)	-7.20	-7.40	-7.56
J_{2s} (10^{-21} J)	-1.12	-0.94	-0.79
J_{3s} (10^{-21} J)	+1.53	+1.49	+1.45

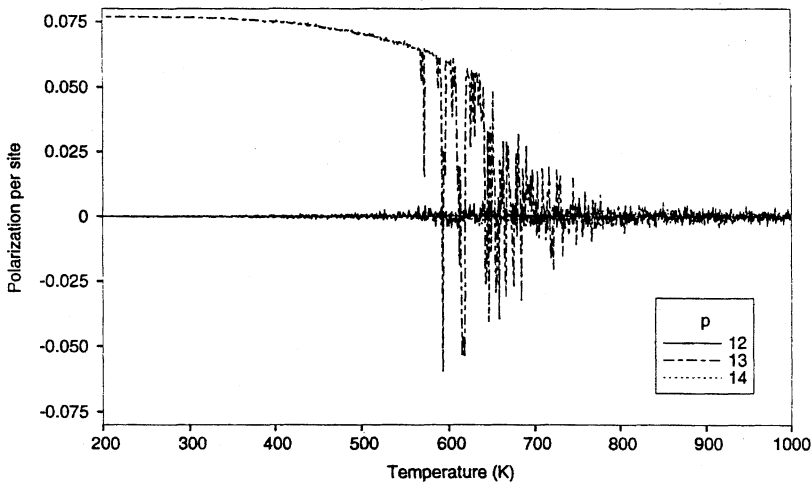


Fig. 8. Polarization per site in 12-B, 13-B and 14-B lattices.

Polarization curves for the B type lattices are shown in Fig. 8. All simulations were started from the low temperature end. The initial configuration was antiferroelectric (Fig. 9) which corresponds to the total relative polarization zero in the 12-B and 14-B and $N_R/(13 \times N_R) = 1/13 = 7.69 \times 10^{-2}$ in the 13-B lattice. As can be seen in Fig. 8 the lattices undergo a transition from the ordered antiferroelectric phase to the disordered paraelectric phase. Because the interaction constants in the B lattice are larger than in the A lattice the transition happens at higher temperatures. Since the two directions of polarization along the MT axis are energetically equivalent the antiferroelectric ground states are doubly degenerate.

7. Effects of an External Electric Field

According to the Ising Hamiltonian (1) the array of dipoles in a MT subjected to an external electric field has the tendency to point in the same direction as the field (to minimize the potential energy due to the interaction with the electric field). Figure 10 shows the effect of an external electric field that points in the positive

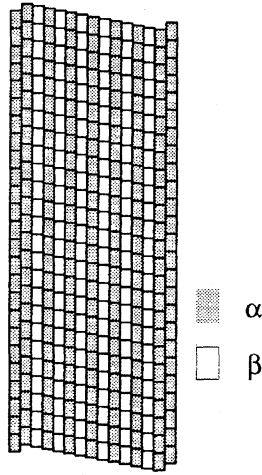


Fig. 9. Antiferroelectric arrangement of dipoles in the 13-B lattice that was used as the initial configuration in the corresponding simulation shown in Fig. 8.

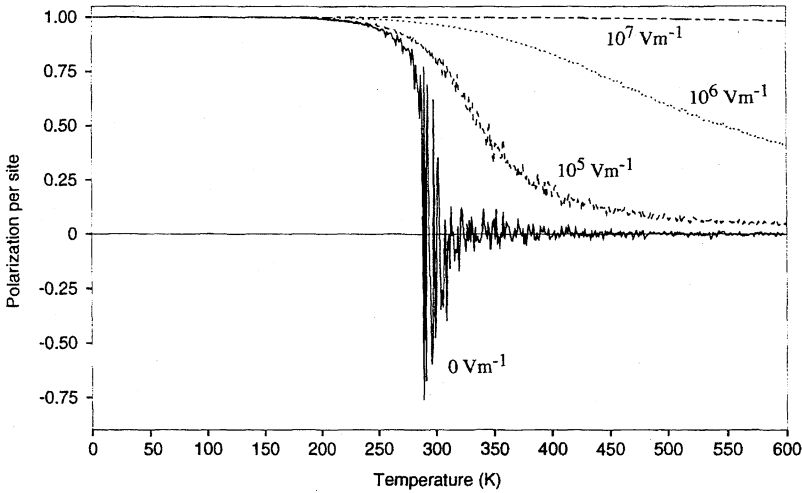


Fig. 10. Polarization curves calculated for a 13-A MT that is subjected to external electric fields 10^5 Vm^{-1} , 10^6 Vm^{-1} and 10^7 Vm^{-1} .

direction along the MT axis on a 13-A MT. To calculate the polarization curves plotted in this figure and also in Figs. 11 and 12 we chose the dipole strength $p = qs = 10^{-27} \text{ Cm}$. As can be seen in Fig. 10 the stronger is the electric field the larger is the number of dipoles that at a temperature T point in the positive direction. This is a consequence of adding the electric potential energy to the dipole-dipole interaction energy. Therefore a larger thermal energy is required to overcome the total potential energy.

The character of the plots in Fig. 10 implies that the electric field starts to be effective at about 10^4 – 10^5 Vm^{-1} . This is because for these fields the potential energy $-pE$ becomes of the order of magnitude of the dipole–dipole interaction energies (see Sec. 3). For $E = 10^7$ Vm^{-1} up to very high temperatures all dipoles point in the direction of the electric field. It can be noted that Vassilev *et al.*²⁶ applied electric pulses with amplitudes of the order of magnitude 10^0 to 10^1 Vm^{-1} to assemblies of MTs with the result that the MTs aligned along the electric field lines. According to our model this would mean that the assembly of dipoles in the tested MTs was in an ordered state so that even small electric fields align the MTs along their field lines. The electric fields that exist inside the cell have been reported to be of similar magnitudes than those used by Vassilev *et al.*²⁷ Hence in our model the electric fields inside the cell wouldn't have an effect on the dipolar ordering in MTs. However, for example, in membranes the electric fields are of the order of 10^6 – 10^7 Vm^{-1} .

In the 12- and 14-protofilament A lattices the effect of an external electric field is more complex than in the 13-protofilament A lattice due to the presence of the seam. As was already shown in these lattices the diagonal interaction along the seam that is the strongest causes the two protofilaments along the discontinuity to be polarized in opposite directions. Consequently, at lower fields it is still more energetically favorable for the dipoles to be aligned in opposite directions in the two protofilaments along the seam. At the same time even though the electric field is not sufficiently strong to flip the dipoles along the seam it is strong enough to flip the dipoles inside the lattice. Hence, for such electric fields at low temperatures all protofilaments are polarized in the direction of the electric field except for one of the two protofilaments that form the seam. When the electric field is sufficiently large so that also the dipoles along the seam are in the state with the lowest potential energy if they point in the same direction as the external electric field, the whole lattice becomes completely ordered at low temperatures.

The effect of an external electric field on a 14-A MT is shown in Fig. 11. For $0 \leq E \leq 3 \times 10^6$ Vm^{-1} at low temperatures all dipoles point in the same direction (positive) as the electric field with the exception of one of the protofilaments along the seam in which all dipoles point in the opposite direction. At some transition temperature the polarization starts to decrease with the increasing temperature. When $E = 5 \times 10^6$ Vm^{-1} at low temperatures the configuration of dipoles is the same as for the lower fields. However, as the thermal energy increases with the increasing temperature, in the protofilament along the seam in which the dipoles are oriented in the negative direction some of them can flip up due to random thermal fluctuations while the dipoles in the rest of the MT point in the positive direction. Consequently, the relative polarization is larger than 12/14. After reaching its maximum the polarization decreases as the temperature increases. The electric fields with the magnitude 6×10^6 Vm^{-1} and higher are sufficiently strong so that at low temperatures all dipoles in the MT point in the same direction as the fields.

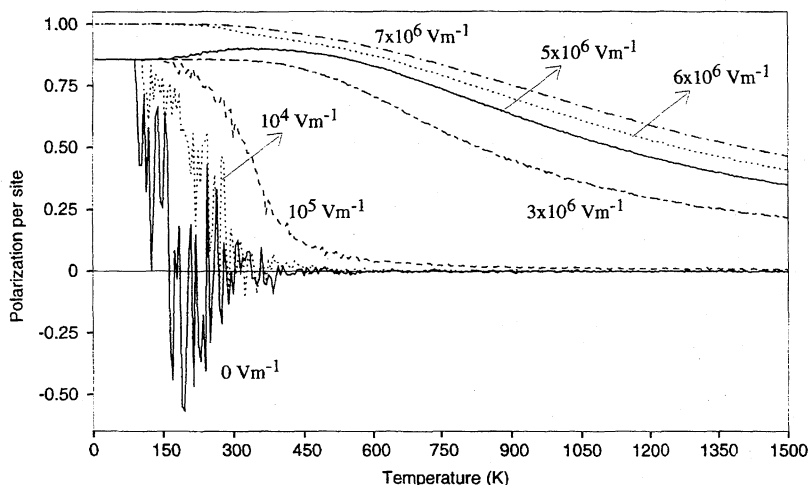


Fig. 11. Polarization per site in a 14-A MT that is subjected to external electric fields. The curves for $E = 3 \times 10^6 \text{ Vm}^{-1}$ and larger fields were calculated starting from random configurations at $T = 2000 \text{ K}$.

The picture would be very similar in the 12-A lattice since its interaction constants do not substantially differ from those in the 14-A lattice (see Table 2).

The behavior of the array of dipoles in the B lattices under the influence of changing external electric field has a similar character as in the 12-A and 14-A lattices and it is shown for a 13-B lattice in Fig. 12. As was discussed in Sec. 6 the ground state of a MT represented by the B lattice is antiferroelectric because the largest interaction constant is J_3 . As can be seen in Fig. 12 when the MT is subjected to lower electric fields, the ground state of the array of dipoles at low temperatures is antiferroelectric. However, as the thermal energy increases with the increasing temperature the random thermal fluctuations allow also the dipoles in the protofilaments with negative polarization to point in the same direction as the electric field. As a result, at fields $5 \times 10^6 \text{ Vm}^{-1}$ to $1.1 \times 10^7 \text{ Vm}^{-1}$ at higher temperatures the polarization per site increases above $1/13$. Once the polarization reaches its maximum given by the combination of the three factors — thermal energy, dipole-dipole interaction energy and the potential energy that is due to the interaction with the electric field — it slowly decreases with the increasing temperature. When the external electric field is larger than a certain value, at low temperatures all dipoles in the lattice point in the direction of the field and at some critical temperature the total polarization starts to decrease. For even larger electric fields the whole lattice would be ordered also at high temperatures.

It can be pointed out that in the 13-B MT due to the odd number of protofilaments the dipoles in the two protofilaments along the seam point in the same direction in the antiferroelectric phase. However, since the values of the interaction constants along the seam are several times smaller than J_3 and J_1 this feature would

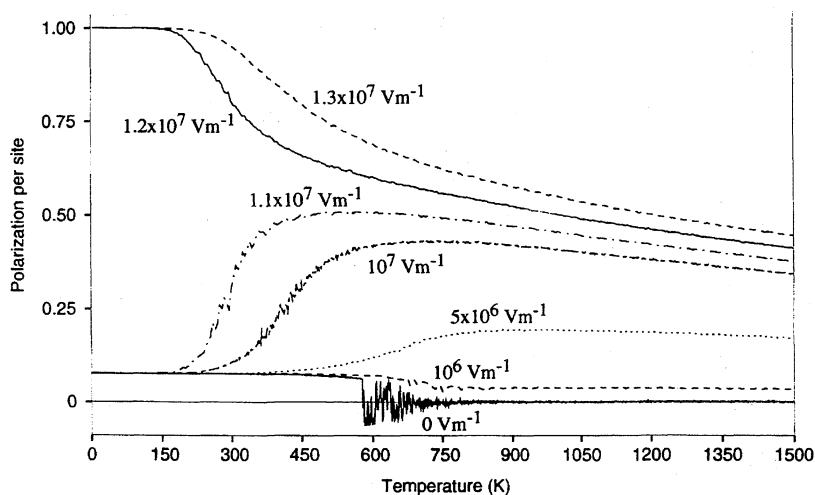


Fig. 12. Polarization per site calculated for a 13-B MT in the presence of external electric fields. The curves for $E = 1.1 \times 10^7 \text{ Vm}^{-1}$ and larger fields were calculated starting from random configurations at $T = 2000 \text{ K}$.

not introduce a substantial difference in the character of the dipolar order in this lattice. Since the values of the interaction constants in the 12-B and 14-B lattices are not very different from those in the 13-B lattice the array of dipoles in MTs with these lattices when they are subjected to external electric fields with various magnitudes will behave in a similar manner. The difference in the magnitudes of the interaction constants will cause different critical temperatures and at low fields and low temperatures the polarization in the B-lattices with even number of protofilaments will be zero.

8. Effects of MAPs

Experimental evidence indicates that MAPs can bind to MTs in both regular and irregular way.⁸ Since a MT is a helical structure it has been suggested that the regular MAP patterns form superhelices.^{8,11} Based on this the influence of MAPs on the dipolar order in the MT lattice was studied here for three different superhelical MAP arrangements. The superhelices were constructed according to¹¹ and they are shown for a 13-A MT in Fig. 13. In the other types of lattices these patterns do not form superhelices due to the presence of the seam (see Figs. 16, 17 and 19). The ratio of the number of MAPs (MAP ratio) to the total number of lattice sites in the three MAP arrangements is 1/11, 1/22 and 1/48. In the simulations these patterns were imposed on MTs 99, 110 and 96 rows long, respectively. Some simulations for MAP ratio 1/11 were performed for a shorter 44 row long MT.

As can be noticed, in forming the patterns we have assumed that each MAP is attached to one tubulin dimer. However, since most MAPs are several times heavier than tubulin they most likely bind to more than one tubulin heterodimer.^{4,28}

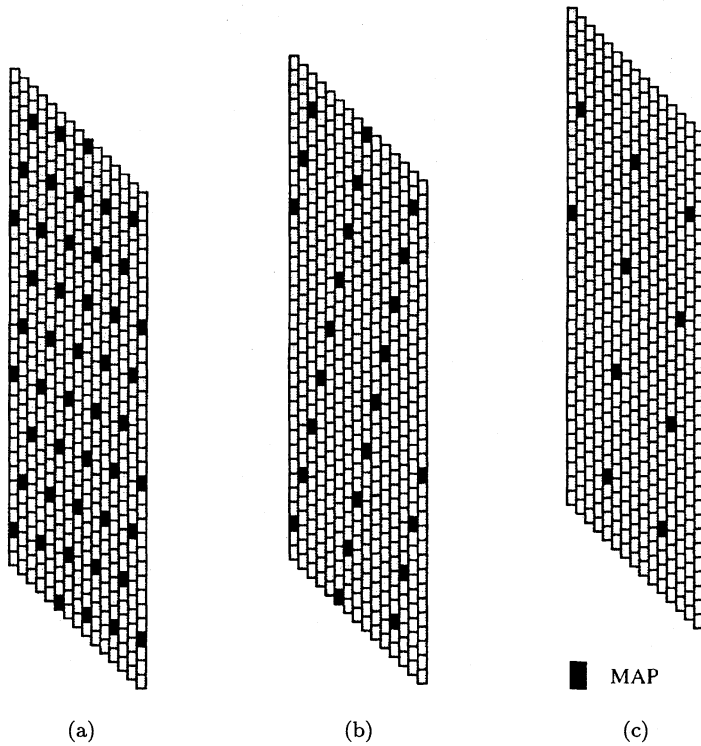
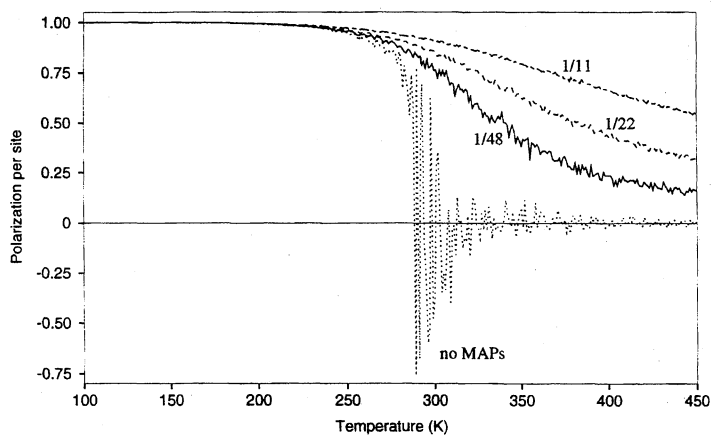


Fig. 13. Three superhelical patterns of MAPs for which the ratios of the number of MAPs to the total number of lattice sites are (a) 1/11, (b) 1/22, (c) 1/48. The patterns are imposed on a 13-protofilament A lattice.

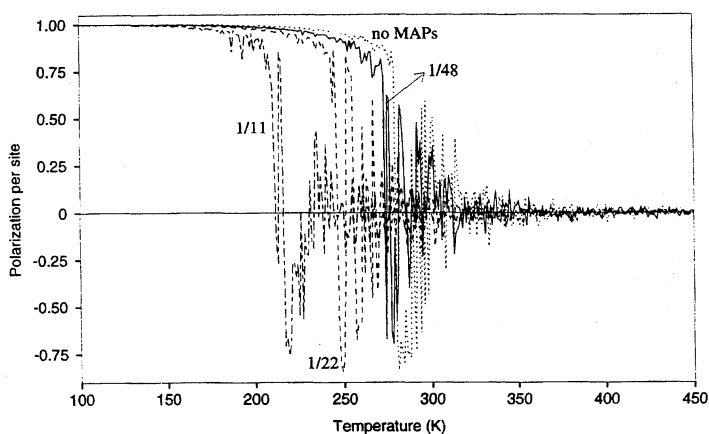
We further assumed that a MAP attached to a tubulin subunit in the MT either prevents the dipole on the tubulin to flip or that it neutralizes the tubulin molecule. Hence, the dipole variable $\sigma_{i,j}$ at the sites of the MT lattice at which MAPs were bound (MAP sites) was set either to one or to zero, respectively.

The polarization curves that represent the dipolar order in a 13-A MT to which MAPs attach according to the three patterns depicted in Fig. 13 are shown in Fig. 14. Figure 14(a) represents a set of calculations in which at the MAP sites $\sigma_{i,j} = 1$. The polarization curves show that the number of tubulin dipoles that point in the same direction, in this case up, is larger in the lattices in which MAPs are present compared to the lattice without MAPs. This behavior is due to the fact that in the 13-A lattice in the state with the lowest energy all dipoles point either up or down (see Sec. 5). If some of the dipoles permanently point up they will force their neighbors also to point up. The more MAPs bind to a MT the greater is at a temperature T the number of dipoles which point in the same direction as the permanent dipoles carried by the tubulins to which the MAPs are attached.

The numerical data obtained for the case when MAPs are represented by $\sigma_{i,j} = 0$ are plotted in Fig. 14(b). The data were normalized with respect to the total number of lattice sites excluding the MAP sites. As can be seen the bound MAPs do not



(a)



(b)

Fig. 14. Polarization curves calculated for a 13-A MT in which MAPs bound to the molecules of tubulin form the three patterns shown in Fig. 13. At the MAP sites $\sigma_{i,j}$ is (a) 1, (b) 0.

change the character of the dipolar order that exists in the lattice in which MAPs are not present — the array of dipoles undergoes a transition from the low temperature ferroelectric phase to the high temperature paraelectric phase. However, the MAPs affect the transition temperature. The larger is the MAP ratio, the lower is the temperature of the transition from the ferroelectric to the paraelectric phase. The reason for this behavior is quite obvious. As was discussed in Sec. 5, in the 13-A lattice the dipolar order is determined by the positive interaction constants J_1 and J_3 . Along the two directions in the MT lattice characterized by these interaction constants in the configuration x , 1 it requires less energy for the dipole that is up to flip down if $x = 0$ compared to the case when $x = 1$. In the configuration x , -1

the dipole that is down flips up at less energy cost for $x = 0$ than for $x = -1$. Thus, if in the starting configuration at low temperatures all dipoles in the MT point in the same direction, it requires less energy and, consequently, lower thermal energy $k_B T$ to undergo a transition to the paraelectric phase if some tubulin molecules in the MT are neutral.

The type of the dipolar ordering in the presence of MAPs in a MT A lattice that has an even number of protofilaments is represented in Fig. 15 that shows a simulation for the 14-A lattice in which at the MAP sites $\sigma_{i,j} = 1$. As in the 13-A lattice the number of dipoles that point up increases with the increasing MAP ratio except for at the seam where the dipoles that interact through the large and negative interaction constant J_{3s} are oriented in opposite directions. Because of the chosen MAP patterns and the type of the boundary conditions the MAPs that are placed in protofilaments 1 and 14 happen to be nearest neighbors along the J_{3s} direction (Fig. 16). Hence, the total relative polarization in the low temperature state is slightly higher than in the case when no MAPs are present in the lattice. Another consequence of this arrangement of MAPs along the seam in the 14-A lattice is that in each of the protofilaments 1 and 14 the direction of polarization can be different in the sections between two nearest MAPs. Consequently, this lattice possesses a multitude of groundstates that are characterized by different configurations of dipoles along the seam.

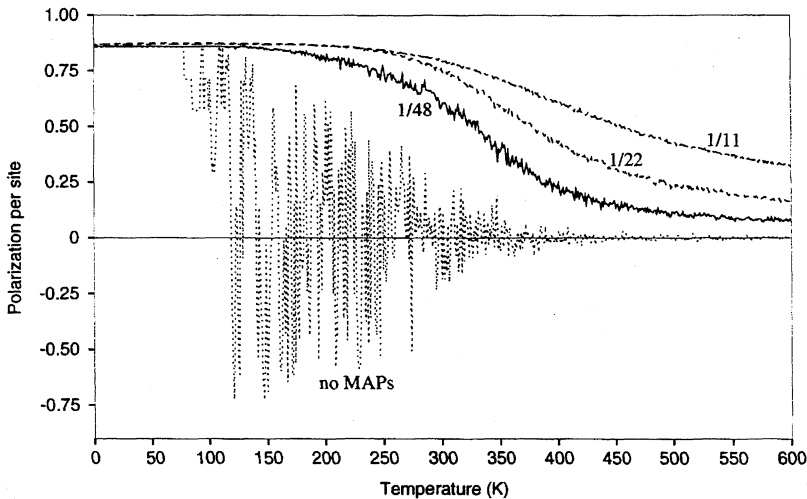


Fig. 15. Polarizations per site in a 14-A MT to which MAPs are bound according to the three patterns shown in Fig. 13. At the MAP sites $\sigma_{i,j} = 1$.

As in the 13-A lattice representing MAPs by $\sigma_{i,j} = 0$ in the 12-A and 14-A lattices does not favor any of their ground states. The dipolar configurations are of a similar type as those found in MTs that do not bind MAPs, however, the

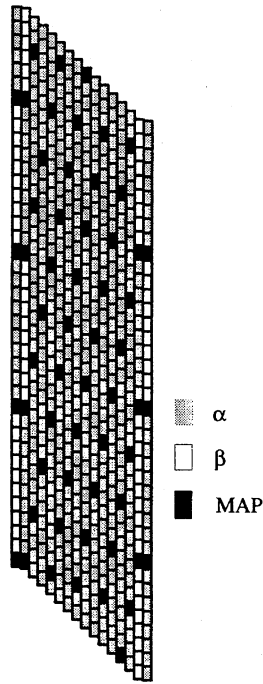


Fig. 16. A section of a 14-A MT that corresponds to the curve for the MAP ratio 1/11 in Fig. 15 at $T = 5$ K.

presence of MAPs introduces also groundstates of different types. We performed numerical calculations for a 14-A MT for all three MAP ratios starting from random configurations at high temperatures. The configuration at which the array of dipoles arrived at low temperatures most often was of the type shown in Fig. 17(a) for the MAP ratio 1/11. It can be seen that along the seam each two dipoles that interact through J_{3s} point in opposite directions. The assembly of dipoles forms two subclusters in each of which the directions of polarization is different [compare to Fig. 6(b)]. The boundary between the two subclusters forms along one of the diagonal lines that connect MAPs since it consumes less energy to flip a dipole in the neighborhood of a tubulin dimer that does not carry any dipole moment.

As was discussed in Sec. 5 the configuration in Fig. 17(a) is not a ground state of the array of dipoles, but it is close to it. Another low temperature state that is close to the state with the lowest energy was obtained for a MT that bound MAPs in the ratio 1/22. In this state all dipoles inside the MT pointed down and all dipoles neighboring along J_{3s} were oriented in opposite directions, but the polarization of dipoles varied between two neighboring MAPs along the protofilaments 1 and 14. In many simulations performed for lower MAP ratios or shorter MTs the MT lattice reached one of the groundstates. Two such groundstates are shown in Figs. 17(b) and (c). Figure 17(b) depicts a section of a 110 rows long MT that is in a groundstate

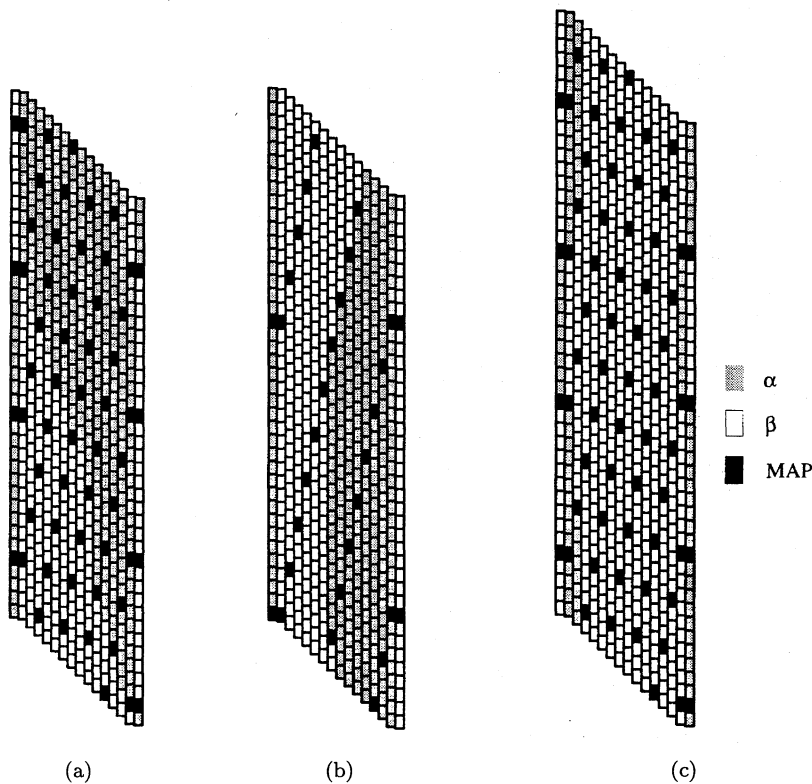


Fig. 17. Three different low temperature dipolar configurations in MTs in the presence of MAPs that are represented by $\sigma_{i,j} = 0$. (a) A section of a MT that is 99 rows long, MAP ratio is 1/11. (b) A section of a 110 row long MT, MAP ratio is 1/22. (c) A 44 row long MT, MAP ratio is 1/11.

for MAP ratio 1/22. Figure 17(c) shows a groundstate in a 44 row long MT to which bind MAPs in the ratio 1/11. The figures imply that there exists a large multitude of groundstates of the types shown. We also found the groundstate configurations that consist of two subclusters, one polarized up and the other down each consisting of 1 up to 13 neighboring protofilaments [see Fig. 6(a)].

Figure 18 shows a set of calculations performed for the 12-B lattice for all three MAP ratios when $\sigma_{i,j} = 1$ at the MAP sites. The simulations were started from the low temperature end. All three lattices are in a ground state or close to a ground state at low temperatures. At $T \sim 200$ K the polarization in the lattice exhibits a maximum, then decreases until it reaches a minimum and eventually increases and at very high temperatures oscillates around the value equal to the MAP ratio. The latter effect cannot be seen because it occurs at temperatures of the order of 10^4 K.

A character of dipolar ordering such as the one represented by Fig. 18 is given by the combination of antiferroelectric interactions, the MAP configuration and thermal energy. At low temperatures at least one of the ground states is such as shown in Fig. 19 for the MAP ratio 1/48. In the background of the antiferroelectric

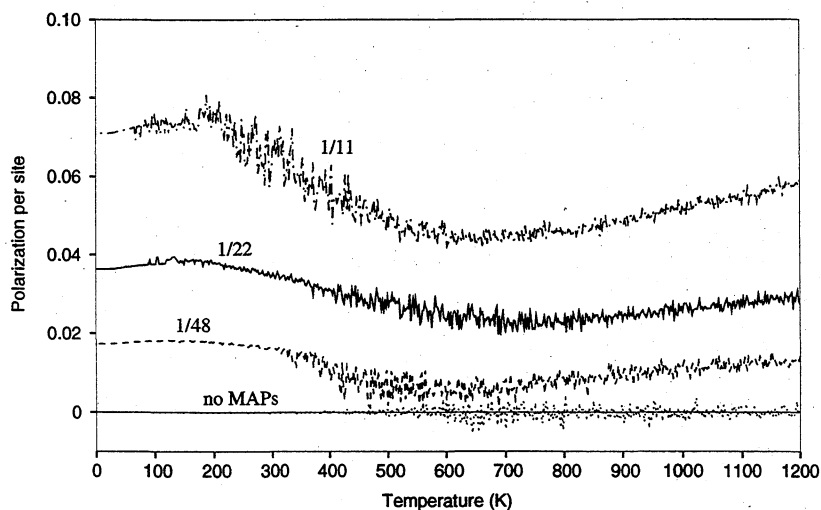


Fig. 18. Polarization curves calculated for the 12-B lattice that contains the three patterns of MAPs. The calculations were started from random configurations at $T = 1500$ K and $\sigma_{i,j} = 1$ at the MAP sites.

arrangement of dipoles are inserted MAPs which at low temperatures do not have an effect on their surrounding dipoles inside the lattice. Along the seam, however, for the dipoles in the 12th row which are neighboring to MAPs in the 11th row it is more energetically favorable to be down. The energy of each of these dipoles is -0.86×10^{-21} J. The relative polarization that corresponds to this state in the lattice is then $20/(12 \times N_R) = 0.0174$. As the temperature increases the thermal fluctuations allow the dipoles along the seam which are neighboring to MAPs to flip into the energetically unfavorable up state ($+0.86 \times 10^{-21}$ J) since the flip does not require much energy. To flip any other dipoles would require several times more energy. Thus, the polarization increases. It should be realized, though, that this increase is due to flipping of only a few dipoles. After the polarization reaches its maximum it decreases because the thermal energy increases and as a consequence some of the dipoles inside the lattice can flip. From its minimum the polarization slowly increases and eventually reaches a state in which all dipoles in the lattice except for at the MAP sites are oriented at random so that the total polarization is equal to the MAP ratio. This happens at temperatures of about 5×10^4 K at which the thermal energy is very large so that it completely overcomes the influence of the dipole-dipole interactions.

For the MAP ratio 1/48 the ground state shown in Fig. 19 was reached at low temperatures in the simulation shown in Fig. 18. When the MAP ratios were 1/22 and 1/11 the states at low temperatures were close to this groundstate. But even for the highest MAP ratio 1/11 the groundstate above existed in shorter MTs (44 rows). We also found that at low temperatures in lattices that contain a smaller

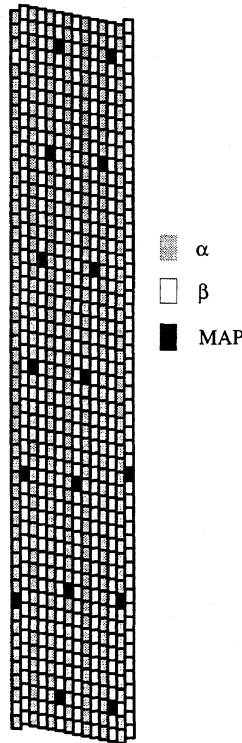


Fig. 19. A ground state in the 12-B lattice that corresponds to $T = 4$ K on the curve for the MAP ratio $1/48$ in Fig. 18.

number of MAPs, so that the spaces between two neighboring MAPs along one protofilament are larger, and for $\sigma_{i,j} = 1$ at the MAP sites, along one protofilament can form subsections in which the dipoles are oriented in opposite directions. Most MAPs in the lattice occur in the subsections in which the dipoles are up to minimize the energy of the interaction along the J_1 direction.

In the 13-B and 14-B lattices the effect of MAPs represented by $\sigma_{i,j} = 1$ is similar as in the 12-B lattice. The dipolar order in the MT is a result of the competition between the thermal energy, dipole-dipole interactions and a MAP configuration. Unlike the lattices that consist of an even number of protofilaments in the 13-B lattice to the antiferroelectric ground state can correspond two different values of polarization: $-1/13$ when all dipoles in six rows point up and all dipoles in seven rows point down, or $+1/13$ in the alternative case. The positive branch of polarization is most likely to appear in lattices that contain higher ratios of MAPs which are represented by $\sigma_{i,j} = 1$. When $\sigma_{i,j} = 0$ at the MAP sites, the lattice is equally likely to choose one of the two branches of polarization. This feature of the 13-protofilament B lattice is an additional factor to those listed above that determine the complex dipolar behavior in the lattices that contain MAPs.

As in the A lattice, if the dipole variable at each MAP site is zero the MAPs do not affect the type of the dipolar ordering in the B lattice, but their presence lowers the transition temperature. In the antiferroelectric phase in the two directions characterized by the large interaction constants J_3 and J_1 it costs less energy to flip a dipole one of whose nearest neighbors is a MAP. Hence, the transition between the antiferroelectric phase and the paraelectric phase takes place at lower temperatures compared to the lattices in which MAPs are not present. To see this, in Fig. 20 is plotted polarization per site calculated for a 13-protofilament MT which contains MAPs in the ratio 1/11 represented by $\sigma_{i,j} = 0$. The simulation started from the high temperature end and the lattice which contains MAPs chose the negative branch of polarization.

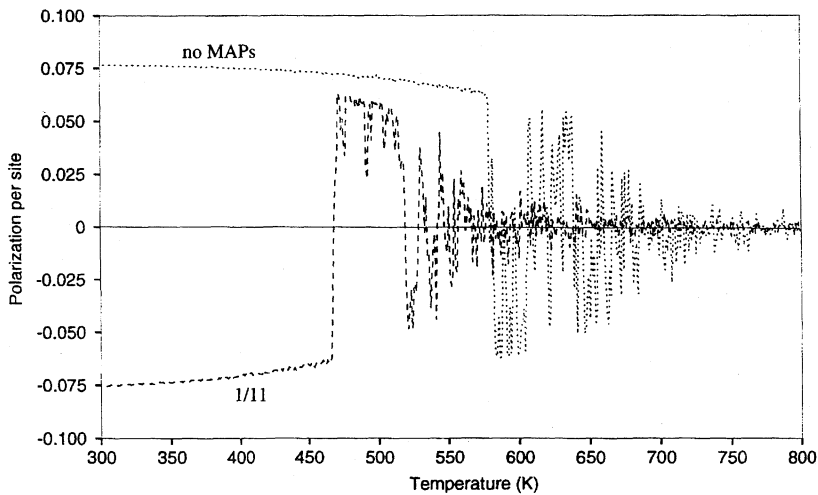


Fig. 20. Relative polarization in the 13-B lattice that contains MAPs represented by $\sigma_{i,j} = 0$ in the ratio 1/11. The calculations were started from a random configuration at $T = 1500$ K.

9. Dipoles Tilted in the β State

To study the dipolar order in lattices in which the dipoles in the β state are tilted by an angle θ with respect to the MT axis we performed calculations for the values of θ in the range $+90^\circ$ to -90° with a step $\pm 10^\circ$ except for the largest angles that were taken to be $\pm 89^\circ$. The sign “+” refers to tilting to the left and “-” refers to tilting to the right with respect to the MT axis. It should be pointed out that tilting to the right and tilting to the left by the same angle are not equivalent in the 12-, 13- and 14-protofilament MT lattices because the distances between two nearest tubulin dimers in the two diagonal directions are not the same (see Table 1, Figs. 3 and 4).

In Fig. 22(a) are plotted curves that represent relative polarization in the 13-A lattice for some “+” values of θ . When $\theta = +10^\circ$ to $+70^\circ$, at low temperatures

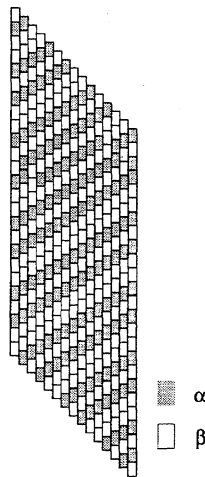


Fig. 21. A section of a MT that corresponds to the curve for $\theta = +89^\circ$ in Fig. 22(a) at $T = 4$ K.

all dipoles are in the tilted state and the value of relative polarization is equal to $(-\cos \theta)$, i.e. it increases with increasing θ . Since $|\cos \theta| < 1$ for $0^\circ < \theta < 90^\circ$, at high temperatures, when due to the thermal energy the dipoles flip into the nontilted state, the polarization per site becomes positive.

The configuration of dipoles for $\theta = +80^\circ$ and $+89^\circ$ at low temperatures is shown in Fig. 21. The dipoles that are in the same state have the tendency to order themselves along the diagonal direction characterized by the constant J_2 and along the J_3 direction the dipole states tend to alternate.

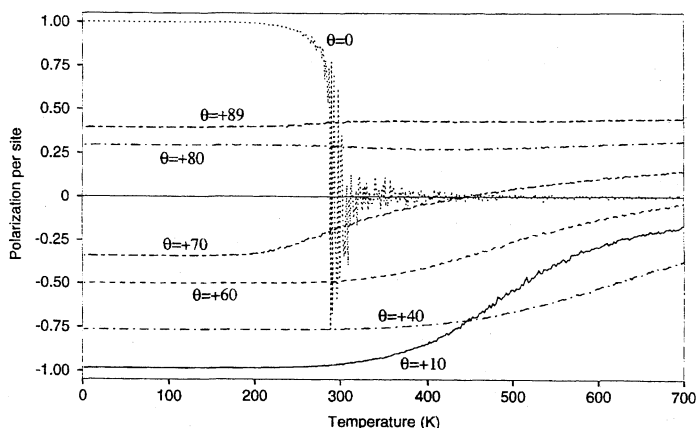
Table 4. The interaction constants in the 13-A lattice for $\theta = -10^\circ, -30^\circ, \pm 40^\circ, +60^\circ, +70^\circ, \pm 80^\circ$.

θ ($^\circ$)	+40	+60	+70	+80	-10	-30	-40	-80
J_1 (10^{-21} J)	-3.07	-1.98	-1.43	-0.94	-4.14	-3.54	-3.07	-0.94
J_2 (10^{-21} J)	-3.97	-5.29	-5.46	-5.32	+2.11	+3.76	+4.14	+2.93
J_3 (10^{-21} J)	+1.33	+1.90	+1.92	+1.80	-2.29	-3.58	-3.99	-3.62

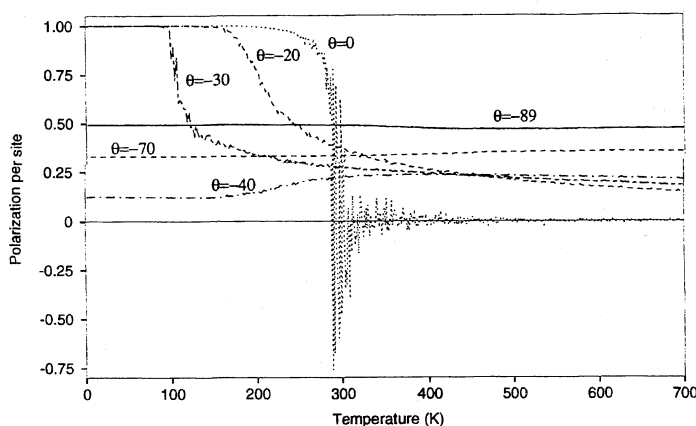
Table 4 which contains the values of the interaction constants J_1 , J_2 and J_3 for certain angles θ shows why the above types of dipolar order arise. For angles less than $+70^\circ$ the constants J_1 and J_2 are negative and larger in magnitude than the positive constant J_3 . Consequently, in the state with the lowest energy all dipoles have the same orientation. Whether they are tilted or nontilted is determined by the second term in the Hamiltonian (5) [the dipolar ordering depends on the difference of energies, see (9)].^c For $\theta = +70^\circ$ the magnitude of the negative J_1 is smaller

^cThe energy of a dipole due to the interaction with its nearest neighbors can be calculated directly from (4) by summing up the appropriate constants J_{nm} .

than the positive J_3 but in the ground state all dipoles are nontilted as a result of the combination of J_n 's, V_{eff} and thermal energy. For $\theta = +80^\circ$, J_3 is sufficiently large compared to the magnitude of J_1 . As a consequence the dipoles mostly are in the same state along the J_2 direction and their states mostly alternate along the J_3 direction. Hence, when θ is large enough and positive the low temperature configuration of dipoles in the 13-A lattice is close to antiferroelectric with respect to the diagonal J_2 direction.



(a)



(b)

Fig. 22. Polarization per site in the 13-A lattice for selected angles θ . The calculations were started from random configurations at $T = 1000$ K. (a) “+” θ 's, (b) “-” θ 's.

Figure 22(b) represents polarizations per site in the 13-A lattice for negative θ 's. At low temperatures for angles up to -30° all dipoles in the MT lattice are nontilted. The ground state for $\theta = -40^\circ$ and larger angles is shown in Fig. 23.

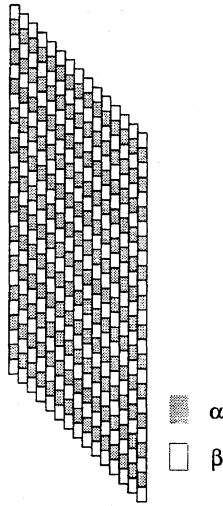


Fig. 23. A section of a MT that corresponds to $T = 4$ K on the curve for $\theta = -40^\circ$ in Fig. 22(b).

The dipoles in the same state form rows along the J_3 direction and the sign of polarization along the J_2 direction alternates. This is a similar kind of order as in the case of $\theta = +80^\circ$ and $+89^\circ$ except for that the types of polarization along the two diagonal directions characterized by the constants J_2 and J_3 are interchanged and this lattice is antiferroelectric with respect to the J_3 direction (see Table 4).

The dipolar ordering in MTs represented by the A lattices with an even number of protofilaments can be expected to be similar to that in the 13-A lattice. Due to the different number of protofilaments the stagger and therefore the interaction energies in these lattices are slightly different. The configuration of dipoles is also influenced by the presence of the seam. Relative polarization in the 14-A lattice for selected positive θ 's is shown in Fig. 24(a). In the low temperature state for $\theta < +40^\circ$ in one of the protofilaments along the seam all dipoles are nontilted and in the rest of the lattice the dipoles are in the tilted state. When θ ranges from $+40^\circ$ to $+70^\circ$ dipoles in the whole lattice are tilted at low temperatures. For angles $+80^\circ$ and $+89^\circ$ in the ground state the configuration of dipoles is antiferroelectric with respect to the direction characterized by J_2 and it is shown in Fig. 25.

In Fig. 24(b) are plotted polarization curves obtained for the 14-A lattice for some negative angles θ . When $-40^\circ < \theta$ the dipoles in the ground state are nontilted except for one of the protofilaments along the seam in which all dipoles are tilted. For $\theta < -40^\circ$ the configuration of dipoles in the ground state is antiferroelectric with respect to the J_3 direction.

To illustrate the influence of tilting of the dipole in the β state on the dipolar ordering in a MT that possesses a B lattice, in Fig. 26 are shown polarizations per site calculated for the 13-B lattice for different angles θ . In agreement with expectations as the tilting of the dipole and therefore the values of the interaction

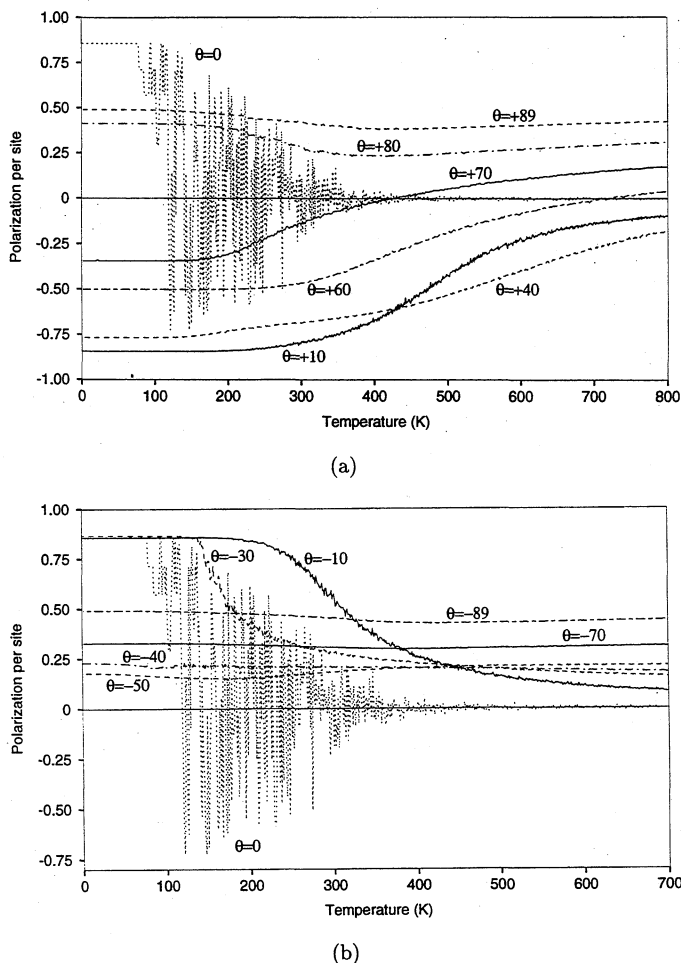


Fig. 24. Polarization per site in the 14-A lattice for selected angles θ . The calculations were started from random configurations at $T = 1000$ K. (a) “+” θ ’s, (b) “-” θ ’s.

constants in the lattice change (Table 5), the type of the low temperature configuration of the dipoles is altered. For angles $0 < \theta \leq +50^\circ$ the configuration in the ground state of the 13-B lattice is antiferroelectric. The dipoles are tilted in odd and nontilted in even protofilaments. When θ is in the range 0 to -30° , the low temperature configuration is also antiferroelectric but the dipolar states in the odd and even protofilaments are interchanged compared to the previous case. In the ground states that correspond to $+60^\circ \leq \theta \leq +90^\circ$ and $-90^\circ < \theta < -40^\circ$ all dipoles in the lattice are tilted.

The effect of external electric fields and MAPs in a MT lattice in which the dipoles are tilted with respect to the longitudinal axis at least in one of their states can be investigated in a similar way as in the case when the dipole in both of its

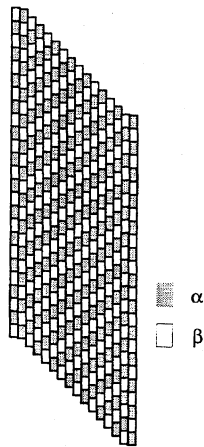


Fig. 25. A section of a MT that corresponds to $T = 4$ K on the curve for $\theta = +89^\circ$ in Fig. 24(a).

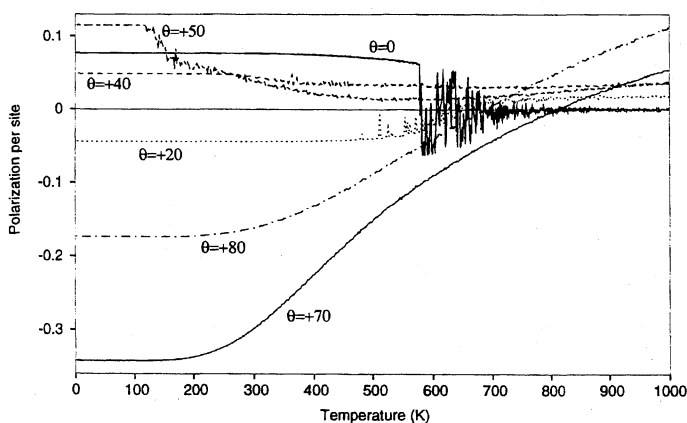
Table 5. The interaction constants in the 13-B lattice for $\theta = -10^\circ, +20^\circ, -30^\circ, -40^\circ, +50^\circ, +60^\circ, \pm 80^\circ$.

θ ($^\circ$)	+20	+50	+60	+80	-10	-30	-40	-80
J_1 (10^{-21} J)	-3.09	-2.54	-1.98	-0.94	-4.14	-3.54	-3.07	-0.94
J_2 (10^{-21} J)	-2.34	-2.60	-2.46	-1.92	-1.23	-0.35	-0.04	+0.78
J_3 (10^{-21} J)	+7.96	+5.47	+4.09	+1.30	+6.41	+3.42	+1.66	-3.78
J_{2s} (10^{-21} J)	-1.67	-4.78	-5.29	-5.32	+2.11	+3.76	+4.14	+2.93
J_{3s} (10^{-21} J)	+0.10	+1.70	+1.90	+1.80	-2.29	-3.58	-3.99	-3.62

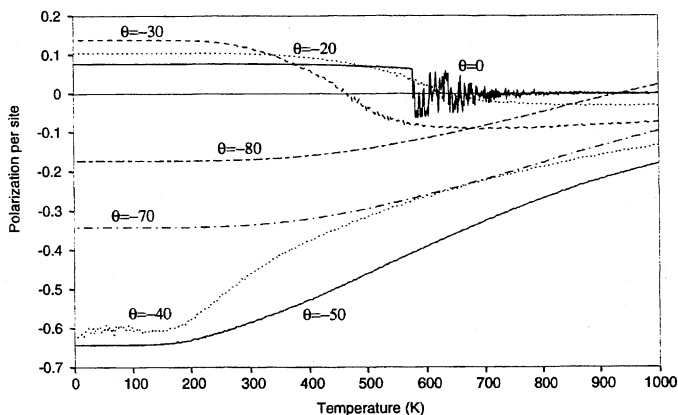
states is nontilted. However, since the tilted and nontilted dipolar states are not energetically equivalent the effect is more complex — it matters in which direction point the external electric fields and whether the dipole states on the tubulin dimers to which bind MAPs are tilted or nontilted.

10. Discussion

In this paper the classical two-dimensional Ising model has been used to investigate the properties of an array of dipoles placed at sites of the hexagonal MT lattice at different temperatures. The lattices of both A and B type for the numbers of MT protofilaments 12, 13 and 14 have been studied. The analysis was supported by numerical simulations using the Monte Carlo method. Each dipole, which represented the charge distribution within one tubulin dimer, was assumed to possess two distinct states whose projections on the MT axis were oriented in opposite directions. The two dipolar states were assumed to be (i) both parallel to the MT axis, (ii) the dipole in one of the states was assumed to be tilted with respect to the MT axis in the two-dimensional space.



(a)



(b)

Fig. 26. Polarization per site in the 13-B lattice for selected angles θ . The calculations were started from random configurations at $T = 1000$ K. (a) "+" θ 's, (b) "-" θ 's.

The investigation has shown that a MT may exist in different dielectric phases at different temperatures. At low temperatures the array of dipoles can form regular patterns. The kind of the pattern depends on the type of the lattice, the number of protofilaments and the angle by which the dipole on the tubulin is tilted. With increasing temperature the regular ordering decays due to increasing thermal energy and at sufficiently high temperatures the dipoles are oriented at random. The range of temperatures over which the transition from the ordered phase to the disordered phase takes place depends on the magnitude and orientation of the dipole moment, relative permittivity of the MT and the geometrical dimensions of the MT lattice.

The dipolar ordering is also affected by the presence of external electric fields and binding of MAPs. According to the model the electric field forces the dipoles

to point in its direction. This means that when a MT is subjected to an electric field it is more likely to be in a ferroelectric phase. The influence of MAPs was studied for three different superhelical patterns that the attached MAPs formed on a MT. We assumed that each MAP binds only to one tubulin dimer and that the associated protein either prevents the dipole on this dimer from flipping or that it neutralizes it. In the first case, naturally, the MAPs enhance the number of dipoles that point in the same direction as the dipole on the MAP tubulin. In the second case the character of the dipolar ordering is the same as in MTs that do not bind any MAPs, however, the presence of neutral tubulin dimers shifts the transition from the ordered to the disordered phase to lower temperatures. If each MAP binds to more than one tubulin molecule these effects can be expected to be greater. In both cases the numerical simulations have shown that the presence of MAPs gives rise to a number of groundstate dipolar configurations that have a more complex character than those found in MTs without MAPs.

The model can be improved by taking into account the curvature of the MT which is a more precise description than the two-dimensional view employed here (see Sec. 3). Then the orientation of the dipole would be determined by two angles in the three-dimensional space. One could also consider the dipole on the tubulin dimer to be tilted in both states by different angles. Introducing these additional parameters would provide more variability to the model. It can be noted that we have made the simplest assumption and took into account only the nearest neighbor interactions between the tubulin dimers. The interactions with the more distant dipoles could be also included but their importance decreases nearly as r^{-3} .

In this work a variety of dipolar configurations has been discussed that can exist in a two-dimensional hexagonal lattice at different temperatures for different parameters of the lattice and different orientation of one of the two dipolar states. The combination of the latter parameters, the dipole length, the charge on the dipole and the permittivity of the MT determines which of these dielectric states might exist in a MT at body temperature. The arrangement of the tubulin heterodimers in a MT is quite well known from experimental measurements. As was discussed in Sec. 3 the charge distribution within a tubulin dimer is not exactly known but it may have dipolar character due to the larger density of charges at one tubulin heterodimer end. Taking the value of the relative permittivity 20 and $q^2 s^2 / \epsilon_r = 12 \times 10^{-56} \text{ C}^2 \text{ m}^2$ gives the dipole strength $qs = 15.5 \times 10^{-28} \text{ Cm}$. Then for $s = 10^{-9} \text{ m}$ the charge on the dipole is about 10 electronic charges and for $s = 2 \times 10^{-9} \text{ m}$ the q is about 5 electronic charges. This means that when the dipole switches its state quite a few charges have to be moving. These assumptions must hold if the states characterized by the regular patterns of dipoles should exist in MTs (see Sec. 5). It should also be pointed out that *in vivo* only a few MTs do not consist of 13 protofilaments (L. A. Amos and W. B. Amos, 1991).

As has been discussed here^{18,19} the tubulin dimers carry permanent dipoles. As was mentioned above the experiments of Vassilev *et al.*²⁶ suggest that MTs are both electric and magnetic structures. Electromagnetic properties of MTs can be the

basis of some of the MT functions. For example, an attractive/repulsive electrostatic interaction between the dipole on free tubulin and the electric field produced by the charges on the tubulin dimers assembled in the MT could play a role in the MT assembly/disassembly. Electromagnetic interaction could be responsible for regular patterns formed by MTs such as the mitotic spindle during cell division.²⁶ The dipolar states existing on the tubulin dimers coupled to tubulin elastic states may give rise to the solitary waves mentioned above. The solitary waves propagate when the dipoles in a MT form regular low temperature arrangements — ferroelectric, antiferroelectric or ferrielectric.¹⁴ Thus according to the investigation presented in this paper the existence of such waves and the possible effects they could have on the MT behavior would depend on the parameters of the MT lattice and the tubulin dipoles. It should be pointed out that the electrostatic properties of the tubulin heterodimer provide means for the MTs to interact with cellular electric fields (including electric fields from other MTs) which may play an important role in controlling the above mentioned biological processes.

In the future using a more precise model of the charge distribution in the tubulin dimer the interactions directly between the charges could be calculated. If the charges can change positions, as we and some other authors^{1-3,11,29} assume, then different electrostatic fields arise. By moving while switching their positions the charges create electric currents and consequently magnetic fields. Hence the investigation of MTs in the direction suggested in this paper could provide very useful insights into the electromagnetic nature of these cytoskeletal structures.

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References

1. J. A. Tuszyński, B. Trpišová, D. Sept, M. V. Sataří, S. Hameroff, 1994; The cell's microtubules: self-organization and information processing properties. In Proceedings of the conference Nonlinear Excitations in Biomolecules, Les Houches School, May 30 to June 4, 1994. ed. M. Peyrard (Springer) 387-404.
2. J. A. Tuszyński, S. R. Hameroff, M. V. Sataří, B. Trpišová and M. L. A. Nip, *J. of Theoretical Biol.* **174**, 371 (1995).
3. J. A. Tuszyński, B. Trpišová and D. Sept., *Neural Network World* **5**, 675 (1995).
4. B. Alberts, D. Bray, J. Lewis, M. Raff, K. Roberts and J. D. Watson, *Molecular Biology of the Cell*, 3rd edn. (Garland Publishing, New York and London, 1994).
5. E. Mandelkow and E.-M. Mandelkow, *Current Opinion Cell Biol.* **2**, 3 (1990).
6. E. Mandelkow and E.-M. Mandelkow, *Current Opinion Struct. Biol.* **4**, 171 (1994).
7. *Guidebook to the Cytoskeletal and Motor Proteins*, eds. T. Kreis and R. Vale (University Press, Oxford, 1993).
8. L. A. Amos and W. B. Amos, *Molecules of the Cytoskeleton*, MacMillan Molecular Biology Series, ed. Ch. J. Skidmore (MacMillan, 1991).
9. S. R. Martin, M. J. Schilstra and P. M. Bayley, *Biophysical J.* **65**, 578 (1993).
10. T. J. Mitchison and M. W. Kirschner, *Nature* **312**, 232 (1984); *Nature* **312**, 237 (1984).

11. S. Hameroff, *Ultimate Computing* (North-Holland, Amsterdam, 1987).
12. Chou Kuo-Chen, Chun-Ting Zhang and G. M. Maggiora, *Biopolymers* **34**, 143 (1994).
13. M. V. Satařić, R. B. Źakula and J. A. Tuszyński, *Phys. Rev.* **E48**, 589 (1993).
14. B. Trpišová and J. A. Tuszyński, *Phys. Rev.* **E55**, 3288 (1997).
15. R. H. Wade, D. Chrétien and D. Job, *J. Mol. Biol.* **212**, 775 (1990).
16. Y.-H. Song and E. Mandelkow, *J. Cell Biol.* **128**, 81 (1995).
17. Y.-H. Song and E. Mandelkow, *Proc. Natl. Acad. Sci.* **90**, 1671 (1993).
18. H. Athenstaedt, *Ann. N. Y. Acad. Sci.* **238**, 68 (1974).
19. D. L. Sackett, talk at the conference Molecular Biophysics of the Cytoskeleton, Banff (August, 1997).
20. J. M. Yeomans, *Statistical Mechanics of Phase Transitions* (Clarendon Press, Oxford, 1992).
21. E. G. Alexov and M. R. Gunner, *Biophysical Journal* **74**, 2075 (1997).
22. H. Bolterauer, J. A. Tuszyński and M. V. Satařić, *Phys. Rev.* **A44**, 1366 (1991).
23. K. Binder and D. W. Heermann, *Monte Carlo Simulation in Statistical Physics* (Springer-Verlag, Berlin, Heidelberg, 1988).
24. G. Burns, *Sol. Stat. Phys.* (Academis Press, 1993).
25. R. K. Pathria, *Statistical Mechanics* (Pergamon Press, 1993).
26. P. M. Vassilev, R. T. Dronzine, M. P. Vassileva and G. A. Georgiev, *Bioscience Rep.* **2** 1025 (1982).
27. L. F. Jaffe and R. Nuccitelli, *Annual Review of Biophysics and Bioengineering* **6**, 445 (1977).
28. *Microtubules*, eds. J. S. Hyams and C. W. Lloyd (Wiley-Liss, New York, 1994).
29. S. Rasmussen, H. Karampurwala, R. Vaidyanath, K. S. Jensen and S. Hameroff, *Physica* **D42**, 428 (1990).